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COMPUTER

CHAPTER 21

Data Encryption and Cryptography

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CHAPTER 21

Data Encryption and Cryptography

Computer Awareness

IN THIS CHAPTER

- Plaintext and ciphertext
- Symmetric encryption
- Public-key cryptography
- Hashing and message digests
- SHA families
- Certificates and PKI

15.Norton and McAfee are popular antivirus software

CRYPTOGRAPHY

Cryptography is the science and art of protecting information by converting it into an unreadable format so that unauthorized persons cannot access it.

Example Original Message: HELLO

Encrypted Message: 4873r@#$

Only authorized users can convert it back to the original message.

BASIC TERMS IN CRYPTOGRAPHY

Sender :The person or system that sends the information.

Recipient (Receiver): The person or system that receives the information.

Plaintext:The original readable information before encryption.

Example: HELLO

Ciphertext:The encrypted unreadable form of plaintext.

Example: 4873r@#$

Encryption: The process of converting plaintext into ciphertext using an encryption algorithm.

Example: HELLO → Encryption → 4873r@#$

Decryption: The process of converting ciphertext back into plaintext using a decryption algorithm.

Example:

4873r@#$ → Decryption → HELLO

TYPES OF ENCRYPTION

There are two main types of encryption:

1.Symmetric Encryption 2.Asymmetric Encryption

SYMMETRIC ENCRYPTION



Symmetric encryption uses the same secret key. Different example ciphertexts depend on the chosen algorithm and key.

In Symmetric Encryption, the same key is used for both encryption and decryption.

The sender and receiver must know and share the same secret key beforehand.

Also Known As

- Secret Key Encryption
- Private Key Encryption

Example

Key = 12345

Sender encrypts the message using key 12345.

Receiver decrypts the message using the same key 12345.

Types of Symmetric Ciphers

A. Stream Cipher

Encrypts data one bit or one byte at a time.

B. Block Cipher

Encrypts data in fixed-size blocks.

Example

A 128-bit block cipher divides plaintext into blocks of 128 bits and encrypts each block separately.

SYMMETRIC ENCRYPTION ALGORITHMS

1. DES (Data Encryption Standard)

Features

- Developed by IBM
- Block size: 64 bits
- Key size: 56 bits
- Block Cipher

Limitation

Not secure today because the key size is too small.

2. Double DES

Double DES applies DES twice, but meet-in-the-middle attacks prevent it from providing the expected doubling of security. Triple DES (3DES) applies DES three times.

DES is performed three times:

Encrypt → Decrypt → Encrypt. Triple DES is historical: NIST withdrew its standard in 2024 and disallows new encryption under its guidance.

4. AES (Advanced Encryption Standard)

Features

- Current international encryption standard
- Considered the "Gold Standard" of encryption
- Rijndael was designed by Joan Daemen and Vincent Rijmen; NIST selected and standardized it as AES.
- Key Sizes:

◦128-bit

◦192-bit

◦256-bit

Uses

- Banking
- Government systems
- Wi-Fi security
- VPNs
- Cloud storage

5. Blowfish

Features

- Fast symmetric encryption algorithm
- Free to use
- Key size: 32 to 448 bits

6. Twofish

Features

- Successor to Blowfish
- Very fast
- Highly secure
- Free to use

ASYMMETRIC ENCRYPTION



Public-key confidentiality: Alice encrypts for Bob, and Bob decrypts. The source examples of HELLO and Call me today are illustrative, not outputs of a specified algorithm.

Asymmetric Encryption uses two different keys:

1.Public Key

2.Private Key

Also Known As

- Public Key Cryptography
- Public Key Encryption

WORKING

Public Key → Encryption

Private Key → Decryption

Both keys belong to the receiver.

Example

Suppose Alice wants to send a message to Bob.

Bob owns:

- Public Key
- Private Key

Alice encrypts the message using Bob's Public Key.

Bob decrypts the message using his Private Key.

Important Note

Rule 1

Message encrypted with Public Key

↓ Can only be decrypted with Private Key

Rule 2

The sender signs using the private signing key.

Others verify the signature using the public verification key.

Digital signing is not a general rule of encrypting a secret message with a private key.

PUBLIC-KEY ALGORITHMS AND RELATED INFRASTRUCTURE

1. RSA (Rivest–Shamir–Adleman)

Features

- Most popular asymmetric algorithm
- Used in:

◦VPNs

◦Email encryption

◦SSL/TLS

◦Digital Signatures

RSA Key Sizes

- 1024-bit (historical; below modern recommended security levels)
- 2048-bit
- 4096-bit (modern systems)

2. Diffie-Hellman Key Exchange

Developed By

- Whitfield Diffie
- Martin Hellman

Purpose

Used to securely exchange cryptographic keys over an insecure network.

Exam Point

Diffie-Hellman is mainly used for key exchange, not direct encryption.

3. DSA (Digital Signature Algorithm)

Used specifically for digital signatures and authentication.

4. ElGamal Algorithm

Public-key cryptographic algorithm used for:

- Encryption
- Digital Signatures

5. PKI (Public Key Infrastructure)

PKI is a framework that manages digital certificates and public keys.

6. DES (symmetric algorithm; not an asymmetric algorithm)

Data Encryption Standard (DES) is a block cipher with a 56-bit key length that has played a significant role in data security.

DES encrypts 64-bit blocks using a 56-bit effective key; it does not generate the key from the plaintext.

DES was developed by IBM and standardized in the United States.

it uses 56 bit key size

HASHING

Hashing is a process that converts data into a fixed-length value called a Hash Value or Hash Code.

Unlike encryption:

Ordinary unkeyed hashing uses no secret key; keyed hash constructions also exist.

One-way process

Original data cannot be recovered from the hash

Example

Message: HELLO

MD5 hash of uppercase HELLO: EB61EEAD90E3B899C6BCBE27AC581660

Uses of Hashing

- Password Storage
- Data Integrity Verification
- Digital Signatures
- Blockchain

MESSAGE DIGEST

A Message Digest is a fixed-size hash value generated from a message.

Purpose

To verify data integrity.

Exam Point

Message Digest Algorithm (MD5)

- Produces 128-bit hash value
- Widely used historically
- No longer considered secure for cryptographic security

SHA (SECURE HASH ALGORITHM)

Developed By NIST (National Institute of Standards and Technology), USA

SHA Family

SHA-0

- 160-bit hash
- Obsolete

SHA-1

- 160-bit hash
- Vulnerable to attacks
- Not recommended

SHA-2

Most widely used secure hash family.

Variants:

- SHA-224
- SHA-256
- SHA-384
- SHA-512

Hash sizes: 224 to 512 bits

Exam Point

SHA-256 is commonly used in modern security systems.

SHA-3

Latest SHA family.

SHA-3 uses a different design from SHA-2; no algorithm can guarantee protection from all future attacks.

PUBLIC KEY CERTIFICATE

A digital certificate binds a public key to an identity through an issuer's signed statement. Trust depends on validating the certificate and issuer.

KEY DIFFERENCES

SYMMETRIC VS ASYMMETRIC ENCRYPTION

Feature	Symmetric Encryption	Asymmetric Encryption
Keys Used	One Key	Two Keys
Speed	Faster	Slower
Security	Depends on algorithm, key size and use	Depends on algorithm, key size and use
Key Sharing	Secret key must be shared securely	Public keys must be distributed and authenticated; private keys stay secret
Examples	AES, DES, Blowfish	RSA, ElGamal (encryption); DSA (signatures only)
Use	Large Data Encryption	Key Exchange, Digital Signatures

MOST IMPORTANT EXAM FACTS

1.AES is the current encryption standard.

2.DES uses a 56-bit key and is now insecure. 3.Triple DES performs DES three times.

4.RSA is the most widely used public-key algorithm.

5.Diffie-Hellman is used for key exchange.

6.DSA is used for digital signatures.

7.MD5 generates a 128-bit hash.

8.SHA-1 produces a 160-bit hash.

9.SHA-256 belongs to SHA-2 family.

10.Hashing is one-way and cannot be decrypted.

11.Public Key encrypts, Private Key decrypts.

12.Digital Certificates are issued by Certificate Authorities (CA).

13.PKI manages public keys and digital certificates.

QUICK REVISION NOTES

- Cryptography = Protection of information.
- Plaintext = Readable data.
- Ciphertext = Encrypted data.
- Encryption = Plaintext → Ciphertext.
- Decryption = Ciphertext → Plaintext.
- Symmetric Encryption = Same key.
- Asymmetric Encryption = Public Key + Private Key.
- AES = Widely used modern symmetric encryption standard.
- RSA = Most popular asymmetric algorithm.
- Diffie-Hellman = Key exchange.
- DSA = Digital Signature.
- Hashing = One-way process.
- MD5 = 128-bit hash.
- SHA-1 = 160-bit hash.
- SHA-256 = Most commonly used SHA version.
- PKI = Manages certificates and public keys.
- Digital certificate = Issuer-signed binding of identity to a public key.

WEB SEARCH ENGINES

A Web Search Engine is a specialized software system or online service that helps users find information on the Internet by entering keywords or phrases.

It searches the World Wide Web (WWW) and returns relevant results such as:

- Web pages

- Images
- Videos
- Documents
- News articles
- Other online resources

The results displayed by a search engine are called Search Results or Hits.